



ABBOTSLEIGH

**2012**  
**HIGHER SCHOOL CERTIFICATE**  
**Assessment 4**

# Mathematics Extension 1

## General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question
- Make sure your HSC candidate Number is on the front cover of each booklet.
- Start a new booklet for Each Question
- Answer the Multiple Choice questions on the answer sheet provided.

Student's Name: \_\_\_\_\_

Student Number: \_\_\_\_\_

Teacher's Name: \_\_\_\_\_

## Total marks – ( 70 )

- Attempt Sections 1 and 2
- All questions are of equal value

Section 1      Pages 2 - 6

### 10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section 2      Pages 7 - 13

### 60 marks

- Attempt Questions 11– 14
- Allow about 1 hr and 45 minutes for this section

**Outcomes to be assessed:****Mathematics****Preliminary :****A student**

- P1 demonstrates confidence in using mathematics to obtain realistic solutions to problems
- P2 provides reasoning to support conclusions which are appropriate to the context
- P3 performs routine arithmetic and algebraic manipulation involving surds, simple rational expressions and trigonometric identities
- P4 chooses and applies appropriate arithmetic, algebraic, graphical, trigonometric and geometric techniques
- P5 understands the concept of a function and the relationship between a function and its graph
- P6 relates the derivative of a function to the slope of its graph
- P7 determines the derivative of a function through routine application of the rules of differentiation
- P8 understands and uses the language and notation of calculus

**HSC :****A student**

- H1 seeks to apply mathematical techniques to problems in a wide range of practical contexts
- H2 constructs arguments to prove and justify results
- H3 manipulates algebraic expressions involving logarithmic and exponential functions
- H4 expresses practical problems in mathematical terms based on simple given models
- H5 applies appropriate techniques from the study of calculus, geometry, probability, trigonometry and series to solve problems
- H6 uses the derivative to determine the features of the graph of a function
- H7 uses the features of a graph to deduce information about the derivative
- H8 uses techniques of integration to calculate areas and volumes
- H9 communicates using mathematical language, notation, diagrams and graphs

**Mathematics Extension 1****Preliminary:****A student**

- PE1 appreciates the role of mathematics in the solution of practical problems
- PE2 uses multi-step deductive reasoning in a variety of contexts
- PE3 solves problems involving permutations and combinations, inequalities, polynomials, circle geometry and parametric representations
- PE4 uses the parametric representation together with differentiation to identify geometric properties of parabolas
- PE5 determines derivatives which require the application of more than one rule of differentiation
- PE6 makes comprehensive use of mathematical language, diagrams and notation for communicating in a wide variety of situations

**HSC :****A student**

- HE1 appreciates interrelationships between ideas drawn from different areas of mathematics
- HE2 uses inductive reasoning in the construction of proofs
- HE3 uses a variety of strategies to investigate mathematical models of situations involving projectiles.
- HE4 uses the relationship between functions, inverse functions and their derivatives
- HE6 determines integrals by reduction to a standard form through a given substitution
- HE7 evaluates mathematical solutions to problems and communicates them in an appropriate form.

## SECTION I

10 marks

Attempt Questions 1 – 10

Use the multiple-choice answer sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample  $2 + 4 =$  (A) 2 (B) 6 (C) 8 (D) 9

(A) ☐ (B) ☒ (C) ☐ (D) ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

(A) ☒ (B) ☒ (C) ☐ (D) ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows.

(A) ☒ (B) ☒ (C) ☐ (D) ☐

*correct*  
↖

---

1. The derivative of  $e^{3x} \cos 3x$  is

(A)  $-3e^{3x} \sin 3x$  (B)  $e^{3x} (\cos 3x - 3 \sin 3x)$

(C)  $-9e^{3x} \sin 3x$  (D)  $3e^{3x} (\cos 3x - \sin 3x)$

2. The solution to  $\frac{x^2 - 9}{x} \geq 0$  is

(A)  $-3 \leq x < 0; x \geq 3$  (B)  $-3 \leq x \leq 3$

(C)  $x \leq -3$  or  $x \geq 3$  (D)  $-3 \leq x \leq 3; x \neq 0$

3. The solution to  $2\sin^2 \theta - \sin \theta = 0$  for  $0 \leq \theta \leq \pi$  is

(A)  $\theta = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi$

(B)  $\theta = 0, \frac{\pi}{6}, \frac{5\pi}{6}, \pi$

(C)  $\theta = \frac{\pi}{3}$  or  $\frac{2\pi}{3}$

(D)  $\theta = \frac{\pi}{6}$  or  $\frac{5\pi}{6}$

4. Given  $f(x) = 2\sec x$  for  $0 \leq x \leq \frac{\pi}{2}$ , then  $f^{-1}(x) =$

(A)  $2\cos^{-1}\left(\frac{1}{x}\right)$

(B)  $2\cos^{-1} x$

(C)  $\cos^{-1}\left(\frac{2}{x}\right)$

(D)  $\cos^{-1}\left(\frac{1}{2x}\right)$

5. The angle between the lines  $2x + y = 4$  and  $x + y = 2$ , to the nearest degree, is

(A)  $18^\circ$

(B)  $25^\circ$

(C)  $45^\circ$

(D)  $72^\circ$

6. Evaluate  $\lim_{x \rightarrow 0} \frac{\sin 3x}{5x}$

(A) 0

(B) Undefined

(C)  $\frac{3}{5}$

(D)  $\frac{5}{3}$

7.  $P(x, y)$  divides  $AB$  externally in the ratio 2: 3 where  $A(-1, 2)$  and  $B(3, 5)$ . Find  $P$ .

(A) (11, 11)

(B)  $\left(\frac{3}{5}, \frac{16}{5}\right)$

(C)  $(-3, -4)$

(D)  $(-9, -4)$

8.  $\int \sec 2x \tan 2x dx =$

(A)  $-\frac{1}{2} \cos^3 2x + c$

(B)  $\frac{1}{2} \tan^2 2x + c$

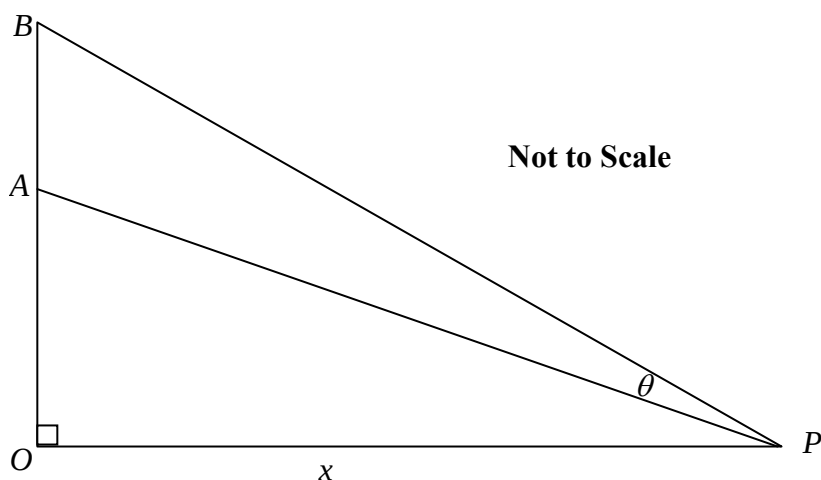
(C)  $\frac{1}{2} \sec 2x + c$

(D)  $\frac{1}{2} \sec^2 2x + c$

9. How many ways can 8 people, each with a different name, be arranged around a circular table if Jill must sit between Jack and Peter?

- (A) 120 (B) 240  
(C) 720 (D) 1440

10. In the diagram below,  $\theta = \angle APB$  and  $OA = 3\text{m}$ ,  $AB = 1\text{m}$ . Which equation is correct?



- (A)  $\theta = \tan^{-1}\left(\frac{1}{x}\right) - \tan^{-1}\left(\frac{3}{x}\right)$  (B)  $\theta = \tan^{-1}\left(\frac{4}{x}\right) - \tan^{-1}\left(\frac{3}{x}\right)$   
(C)  $\theta = \tan^{-1}\left(\frac{1}{\sqrt{9-x^2}}\right)$  (D)  $\theta = \tan^{-1}(3-x)$

**End of Section I**

## SECTION 2

**Total Marks – 60**

**Attempt Questions 11-14**

**All questions are of equal value**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

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**Question 11** (15 marks) Use a SEPARATE writing booklet.

**Marks**

(a) Express  $\frac{{}^{10}C_k}{{}^9C_k}$  in simplest form. **2**

(b) A tennis team of 4 men and 4 women is to be selected from 6 men and 7 women.

(i) Find the number of ways in which this can be done. **1**

(ii) It was decided that two particular women must be selected together or not selected at all. How many teams could be selected in these circumstances? **2**

(c) Find  $\int \cos^2 3x \, dx$  **2**

(d) Using the substitution  $u = \log_e x$ , evaluate  $\int_1^e \frac{\log_e x}{3x} \, dx$  **3**

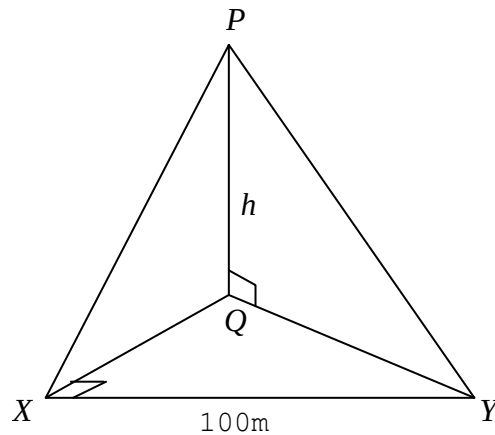
(e) Find the term independent of  $x$  in the expansion of  $\left(x - \frac{1}{2x^3}\right)^{20}$  **3**

(f) Given the roots of  $x^3 - x^2 + 4x - 2 = 0$  are  $\alpha, \beta$  and  $\gamma$ , evaluate

(i)  $\alpha\beta\gamma$  **1**

(ii)  $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ . **1**

**End of Question 11**



- (a) The angle of elevation of a tower,  $PQ$ , from a point  $X$  due south of it is observed to be  $70^\circ$ . From another point  $Y$ , due east of  $X$ , the angle of elevation is  $58^\circ$ .

Given  $XY = 100\text{ m}$ , find the height  $h$  of the tower  $PQ$  to the nearest metre.

**3**

- (b) Consider the function  $f(x) = x - 3 + \log_e x$ .

- (i) Show that there is a root between  $x = 1$  and  $x = 3$

**1**

- (ii) Using the first approximation of the root  $x_1 = 2$ , use Newton's method to estimate the **second approximation** of this root.

Round your answer to 2 decimal places.

**2**

- (c) For the expansion of  $(1+3x)^p(1-2x)^q$  find an expression for the coefficient of  $x^2$ .

**2**

**Question 12 continues on next page**

(d) Consider the function  $y = x \cos^{-1} x$

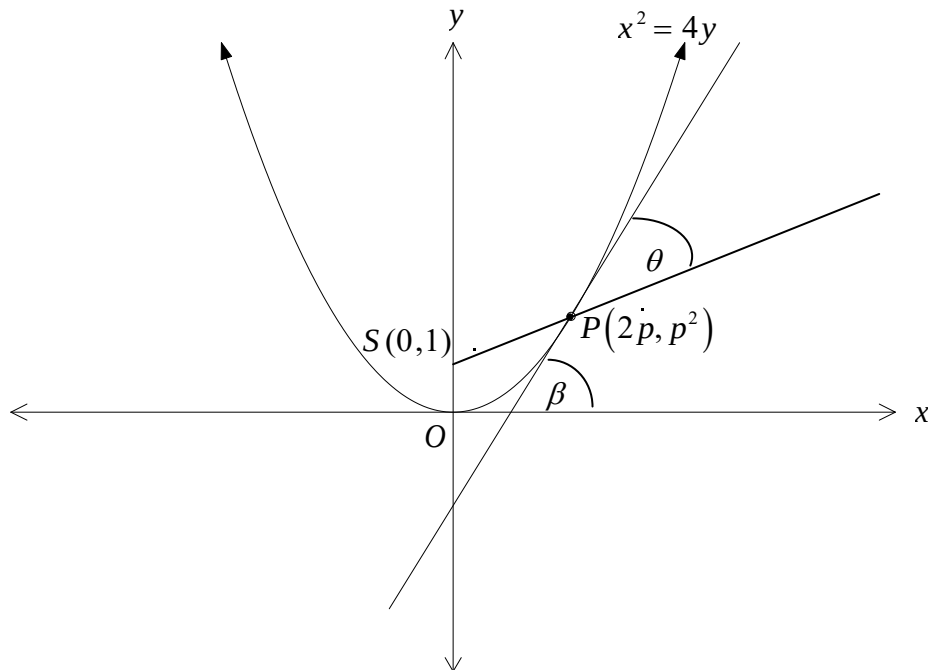
(i) Find  $\frac{dy}{dx}$  **2**

(ii) Hence, or otherwise, evaluate  $\int_0^{\frac{\sqrt{3}}{2}} \cos^{-1} x \, dx$  **3**

(e) Using  $\tan \frac{x}{2} = t$ , or otherwise, prove that  $\frac{2}{1 + \cos x} = \sec^2 \left( \frac{x}{2} \right)$  **2**

**End of Question 12**

(a)



Let  $P(2p, p^2)$  be a point on the parabola  $x^2 = 4y$ , and let  $S$  be the focal point  $(0, 1)$ .

The tangent to the parabola makes an angle of  $\beta$  with the  $x$ -axis.

The angle between  $SP$  and the tangent is  $\theta$ . Assume  $p > 0$  as indicated.

- |       |                                                                             |          |
|-------|-----------------------------------------------------------------------------|----------|
| (i)   | Show that $\tan \beta = p$                                                  | <b>1</b> |
| (ii)  | Show that the gradient of $SP$ is $\frac{1}{2}\left(p - \frac{1}{p}\right)$ | <b>1</b> |
| (iii) | Show that $\tan \theta = \frac{1}{p}$                                       | <b>2</b> |
| (iv)  | Show that the value of $\theta + \beta$ is $\frac{\pi}{2}$                  | <b>2</b> |
| (v)   | Find the coordinates of $P$ when $\theta = \beta$                           | <b>1</b> |

**Question 13 continues on the next page**

(b) Let  $P(x) = x^3 + ax^2 + bx - 18$

Find the values of  $a$  and  $b$  if  $(x+2)$  is a factor of  $P(x)$ , and  $-24$  is the remainder when  $P(x)$  is divided by  $(x-1)$ .

3

(c) (i) Show that  $\sqrt{3} \sin x + \cos x = 2 \sin\left(x + \frac{\pi}{6}\right)$

2

(ii) Hence or otherwise sketch at least **one period** of the graph of  $y = \sqrt{3} \sin x + \cos x$ . Clearly show all  $x$ -intercepts and the amplitude.  
(Note: One period = one complete wave length)

2

(iii) Using your graph, or otherwise, find the general solution of the equation  $\sqrt{3} \sin x + \cos x = 0$

1

**End of Question 13**

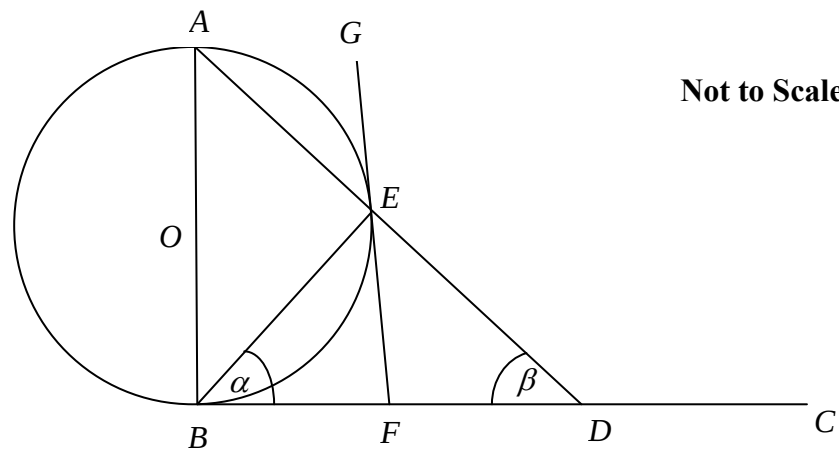
**Question 14** (15 marks) Use a SEPARATE writing booklet.

**Marks**

(a) Find the exact value of  $\cos\left(\sin^{-1}\frac{3}{4}\right)$

**1**

- (b) In the diagram below,  $AB$  is the diameter of the circle, centre  $O$ , and  $BC$  is tangential to the circle at  $B$ . The line  $AD$  intersects the circle at  $E$  and  $BC$  at  $D$ . The tangent to the circle at  $E$  intersects  $BC$  at  $F$ .  
Let  $\angle EBF = \alpha$  and  $\angle EDF = \beta$ .



- (i) Copy the diagram into your writing booklet.  
(ii) Prove that  $\angle FED = \frac{\pi}{2} - \alpha$   
(iii) Prove that  $BF = FD$

**2**

**2**

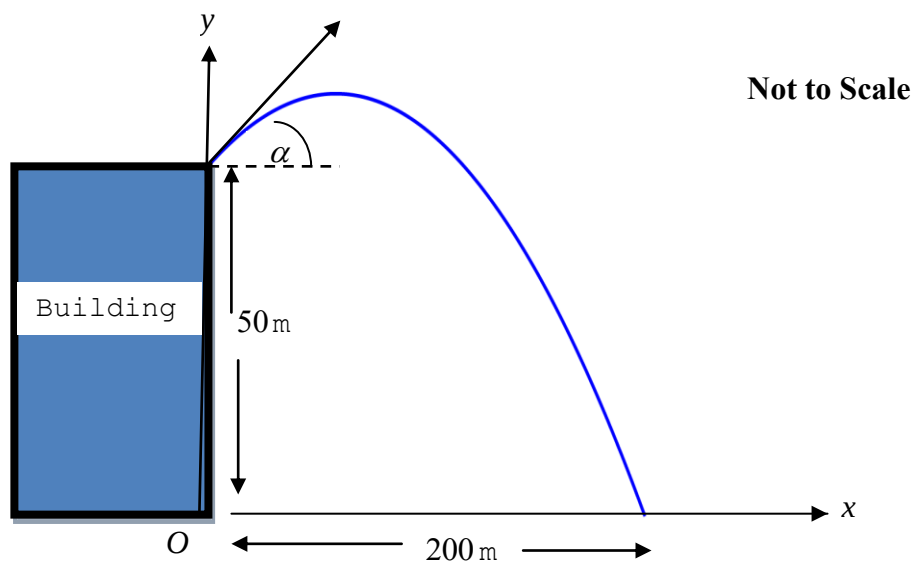
- (c) Use mathematical induction to prove that, for all positive integers  $n \geq 1$ ,

$$3 \cdot 2^2 + 3^2 \cdot 2^3 + 3^3 \cdot 2^4 + \dots + 3^n \cdot 2^{n+1} = \frac{12}{5} (6^n - 1)$$

**3**

**Question 14 continues on next page**

(d)



The diagram shows the path of a projectile launched at an angle of elevation,  $\alpha$ , with an initial speed of 40m/s from the top of a 50 metre high building and lands on the ground 200 metres from the base of the building. The acceleration due to gravity is assumed to be  $10\text{m/s}^2$ .

- (i) Given  $\frac{d^2x}{dt^2} = 0$  and  $\frac{d^2y}{dt^2} = -10$ , show that  
 $x = 40t \cos \alpha$  and  $y = -5t^2 + 40t \sin \alpha + 50$ , where  $x$  and  $y$  are the horizontal and vertical displacements of the projectile in metres from  $O$  at time  $t$  seconds after launching. 2
- (ii) If the projectile is launched at an angle of elevation of  $45^\circ$ , find the maximum height it reaches above the ground. 2
- (iii) Find two possible values for  $\alpha$ .  
Give your answers to the nearest degree. 3

**END OF PAPER**

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**FORMULA SHEET ON BACK**

## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1, x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

NOTE :  $\ln x = \log_e x, x > 0$

Section 1

1. C
2. A
3. B
4. C
5. A
6. D
7. D
8. C
9. B
10. B

Working

$$1. \lim_{x \rightarrow 0} \frac{\sin 3x}{5x} = \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \cdot \frac{3}{5}$$

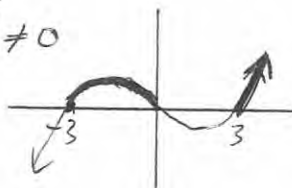
$$= \frac{3}{5} \times 1$$

$$= \frac{3}{5}$$

$$2. x(x^2 - 9) \geq 0, x \neq 0$$

$$x(x-3)(x+3) \geq 0$$

$$-3 \leq x < 0, x \geq 3$$



$$3. \sin \theta (2 \sin \theta - 1) = 0$$

$$\sin \theta = 0 \text{ OR } \sin \theta = \frac{1}{2} \quad (0 \leq \theta \leq \pi)$$

$$\theta = 0, \pi$$

$$\theta = \frac{\pi}{6}, \pi - \frac{\pi}{6}$$

$$\therefore \theta = 0, \frac{\pi}{6}, \frac{5\pi}{6}, \pi$$

$$4. y = 2 \sec x \quad \therefore f^{-1}: x = 2 \sec y$$

$$\therefore \cos y = \frac{2}{x} \quad x = \frac{2}{\cos y}$$

$$y = \cos^{-1}\left(\frac{2}{x}\right)$$

$$5. m_1 = -2 \quad m_2 = -1$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \quad \therefore \theta = \tan^{-1}\left(\frac{1}{3}\right)$$

$$= \left| \frac{-2+1}{1+2} \right|$$

$$= \frac{1}{3}$$

$$6. y = e^{3x} \cos 3x$$

$$y' = \cos 3x \cdot 3e^{3x} + e^{3x} \cdot -3 \sin 3x$$

$$= 3e^{3x} (\cos 3x - \sin 3x)$$

$$7. A(-1, 2) \quad B(3, 5)$$

$$-2; 3$$

$$P = \left( \frac{3x-1+3x-2}{-2+3}, \frac{3x2+5x-2}{-2+3} \right)$$

$$= (-9, -4)$$

8. From table of standard integrals, 7th integral down pag. where  $a=2$

9. Place Jill, then 2 ways of placing Jack + Peter. Then 5! ways to place others  
so  $2 \times 5! = 240$

$$10. \text{ In } \triangle BOP, \tan \angle BPO = \frac{4}{x}$$

$$\therefore \angle BPO = \tan^{-1}\left(\frac{4}{x}\right)$$

$$\text{ In } \triangle AOP, \tan \angle APO = \frac{3}{x}$$

$$\therefore \angle APO = \tan^{-1}\left(\frac{3}{x}\right)$$

$$\theta = \angle BPO - \angle APO$$

$$= \tan^{-1}\left(\frac{4}{x}\right) - \tan^{-1}\left(\frac{3}{x}\right)$$

SECTION 2

$$11. (a) \frac{{}^{10}C_k}{{}^9C_k} = \frac{10!}{k!(10-k)!} \div \frac{9!}{k!(9-k)!}$$

$$= \frac{10!}{k!(10-k)(9-k)!} \times \frac{k!(9-k)!}{9!}$$

$$= \frac{10}{10-k}$$

$$(b) \text{ i) } \frac{5!}{2!} = 60$$

$$\text{ (ii) } \frac{3 \times 4 \times 3 \times 2 \times 1}{2!} = 36$$

$$11.(c) \int \cos^2 3x \, dx$$

$$= \frac{1}{2} \int (1 + \cos 6x) \, dx$$

$$= \frac{1}{2} \left( x + \frac{1}{6} \sin 6x \right) + c$$

$$(d) \int_1^3 \frac{\log_e x}{3x} \, dx$$

$$u = \log_e x$$

$$du = \frac{1}{x} dx$$

$$\text{When } x=3, u = \ln 3$$

$$\text{When } x=1, u = 0$$

$$= \int_0^{\ln 3} \frac{u}{3} \, du$$

$$= \left[ \frac{u^2}{6} \right]_0^{\ln 3}$$

$$= \frac{(\ln 3)^2}{6}$$

$$(e) \text{ General term } {}^{20}C_k x^{20-k} \left(-\frac{1}{2x^3}\right)^k$$

$$= {}^{20}C_k x^{20-k} \times x^{-3k} \times \left(-\frac{1}{2}\right)^k$$

independent of  $x$  means  $x^0$

$$\therefore 20 - k - 3k = 0$$

$$k = 5$$

$$\therefore \text{ term is } {}^{20}C_5 \left(-\frac{1}{2}\right)^5 = -484.5$$

$$(f) (i) -\frac{d}{a} = \frac{-2}{1} = 2$$

$$(ii) \frac{\beta x + \alpha x + \alpha \beta}{\alpha \beta x} = \frac{\frac{4}{1}}{2}$$

$$= 2$$

$$12(a) \text{ In } \triangle PQX, \tan 70^\circ = \frac{h}{xQ}$$

$$\therefore xQ = \frac{h}{\tan 70^\circ}$$

$$\text{In } \triangle PQY, \tan 58^\circ = \frac{h}{yQ}$$

$$\therefore yQ = \frac{h}{\tan 58^\circ}$$

$$QY^2 = QX^2 + 100^2 \text{ since } \angle QXY = 90^\circ$$

$$12(a) \text{ (Continued)}$$

$$\therefore 100^2 = \left( \frac{h}{\tan 58^\circ} \right)^2 - \left( \frac{h}{\tan 70^\circ} \right)^2$$

$$10000 = h^2 \left( \frac{1}{\tan^2 58^\circ} - \frac{1}{\tan^2 70^\circ} \right)$$

$$h^2 = \frac{10000}{0.257987375 \dots}$$

$$h = \sqrt{38761.58661 \dots}$$

$$= 196.879 \dots$$

$$= 197 \text{ m (to nearest metre)}$$

$$(b) f(x) = x - 3 + \log_e x$$

$$(i) f(1) = 1 - 3 + 0$$

$$= -2 < 0$$

$$f(3) = 3 - 3 + \log_e 3$$

$$= \log_e 3 > 0$$

$\therefore$  root lies b/w 1 & 3 as function changes sign

$$(ii) f'(x) = 1 + \frac{1}{x}$$

$$\text{Let } x_1 = 2$$

$$\therefore x_2 = 2 - \frac{f(2)}{f'(2)}$$

$$= 2 - \frac{(2 - 3 + \ln 2)}{1 + \frac{1}{2}}$$

$$= 2 + 0.204 \dots$$

$$= 2.20 \text{ to 2 d.p.}$$

$$(c) (1+3x)(5-2x)^P$$

$$= (1+3x)({}^P C_0 5^P (-2x)^0 + {}^P C_1 5^{P-1} (-2x)^1)$$

$$= (1+3x)({}^P C_0 5^P - 2 {}^P C_1 5^{P-1} x + 4 {}^P C_2 5^{P-2} x^2)$$

$$\therefore \text{coeff of } x^2 \text{ is } 4 {}^P C_2 5^{P-2} - 6 {}^P C_1 5^P$$

12(d)  $y = x \cos^{-1} x$

(i)  $\frac{dy}{dx} = \cos^{-1} x \cdot 1 + x \cdot \frac{-1}{\sqrt{1-x^2}}$   
 $= \cos^{-1} x - \frac{x}{\sqrt{1-x^2}}$

(ii) From (i) :

$$\int \cos^{-1} x - \frac{x}{\sqrt{1-x^2}} = x \cos^{-1} x$$

$$\therefore \int_0^{\frac{\sqrt{3}}{2}} \cos^{-1} x - \frac{x}{\sqrt{1-x^2}} dx = \left[ x \cos^{-1} x \right]_0^{\frac{\sqrt{3}}{2}} + \int_0^{\frac{\sqrt{3}}{2}} \frac{x}{\sqrt{1-x^2}} dx$$

$$= \left( \frac{\sqrt{3}}{2} \cos^{-1} \frac{\sqrt{3}}{2} - 0 \right) + \int_0^{\frac{\sqrt{3}}{2}} \frac{x}{\sqrt{1-x^2}} dx$$

$$= \frac{\sqrt{3}}{2} \times \frac{\pi}{6} - \frac{1}{2} \int_0^{\frac{\sqrt{3}}{2}} 2x (1-x^2)^{-\frac{1}{2}} dx$$

$$= \frac{\pi\sqrt{3}}{12} - \frac{1}{2} \left[ \frac{(1-x^2)^{\frac{1}{2}}}{-\frac{1}{2}} \right]_0^{\frac{\sqrt{3}}{2}}$$

$$= \frac{\pi\sqrt{3}}{12} - \left[ \left(1 - \frac{3}{4}\right)^{\frac{1}{2}} - (1)^{\frac{1}{2}} \right]$$

$$= \frac{\pi\sqrt{3}}{12} - \frac{1}{2} + 1$$

$$= \frac{\pi\sqrt{3} + 6}{12}$$

(e) Using  $\tan \frac{x}{2} = t$ ,

$$\text{LHS} = \frac{2}{1+\cos x}$$

$$= \frac{2}{1 + \frac{1-t^2}{1+t^2}}$$

$$= \frac{2(1+t^2)}{1+t^2+1-t^2}$$

$$= 1+t^2$$

$$= 1 + \tan^2 \left( \frac{x}{2} \right)$$

$$= \sec^2 \left( \frac{x}{2} \right)$$

$$= \text{RHS}$$

12(e) (alt. method)

let  $x = 2\theta$

$$\therefore \text{LHS} = \frac{2}{1+\cos 2\theta}$$

$$= \frac{2}{1+2\cos^2 \theta - 1}$$

$$= \frac{2}{2\cos^2 \theta}$$

$$= \sec^2 \theta$$

$$= \sec^2 \frac{x}{2}$$

$$= \text{RHS}$$

13.(a)  $x^2 = 4y$      $P(2p, p^2)$   
 $S(0, 1)$

(i)  $y = \frac{x^2}{4}$   
 $\frac{dy}{dx} = \frac{x}{2}$

$\therefore$  grad. of tangent at  $P = \frac{2p}{2} = p$

Since  $\beta = \angle$  b/wn tangent +  $x$ -axis

$\tan \beta = \text{grad. of tangent at } P$   
 $= p$

(ii)  $m_{sp} = \frac{p^2 - 1}{2p - 0}$   
 $= \frac{p^2}{2p} - \frac{1}{2p}$   
 $= \frac{1}{2} \left( p - \frac{1}{p} \right)$

(iii)  $\theta = \text{angle b/wn } SP \text{ + tang. at } P$

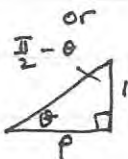
$\therefore m_1 = \frac{1}{2} \left( p - \frac{1}{p} \right)$      $m_2 = p$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{\frac{1}{2}p - \frac{1}{2p} - p}{1 + \frac{p^2}{2} - \frac{1}{2}} \right|$$

$$= \left| \frac{-\frac{1}{2p}(1+p^2)}{\frac{1}{2}(p^2+1)} \right| = \frac{1}{p}$$

13(a) (Continued)

(iv)  $\tan \theta = \frac{1}{p}$   $\tan \beta = p$  

$\therefore \cot \theta = p = \tan \beta$

$\therefore \theta + \beta = \frac{\pi}{2}$  (complementary angles)

OR  $\tan(\theta + \beta) = \frac{\tan \theta + \tan \beta}{1 - \tan \theta \tan \beta}$   
 $= \frac{\frac{1}{p} + p}{1 - 1}$   
 $= \text{undefined}$

$\therefore \theta + \beta = \frac{\pi}{2}$

(v) When  $\theta = \beta$   $\frac{1}{p} = p$   
 $\therefore p = 1$ ,  $P = (2, 1)$

b)  $P(x) = x^3 + ax^2 + bx - 18$

$P(-2) = 0 \therefore -8 + 4a - 2b - 18 = 0$

$P(1) = -24 \therefore 1 + a + b - 18 = -24$

$4a - 2b = 26$  (1)

$a + b = -7$  (2)

$2a - b = 13$  (3)

(1) + (3)  $3a = 6$

$a = 2$

$b = -9$

$\sqrt{3} \sin x + \cos x \equiv R \sin(x + \alpha)$

$R \sin x \cos \alpha + R \cos x \sin \alpha$

$\therefore \sqrt{3} = R \cos \alpha$

$1 = R \sin \alpha$

$R^2 \cos^2 \alpha + R^2 \sin^2 \alpha = 3 + 1 = 4$

$R^2 (\sin^2 \alpha + \cos^2 \alpha) = 4$

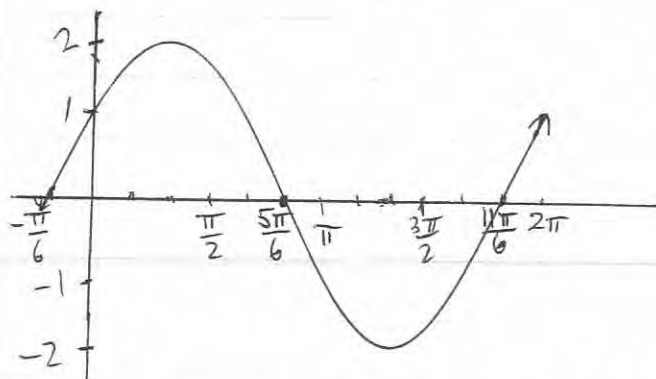
$R = 2$

$\tan \alpha = \frac{R \sin \alpha}{R \cos \alpha} = \frac{1}{\sqrt{3}} \therefore \alpha = \frac{\pi}{6}$

$\sqrt{3} \sin x + \cos x \equiv 2 \sin(x + \frac{\pi}{6})$

13(c) (Continued)

(ii)  $y = 2 \sin(x + \frac{\pi}{6})$

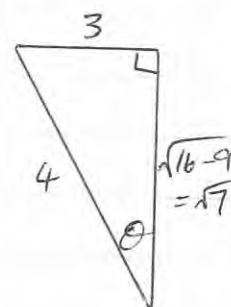


(iii)  $x = k\pi - \frac{\pi}{6}$ ,  $k = \text{any integer}$

OR  $x = \frac{5\pi}{6} + k\pi$ ,  $k = \text{any integer}$

14. (a) Let  $\theta = \sin^{-1} \frac{3}{4}$   
 $\therefore \sin \theta = \frac{3}{4}$

$\therefore \cos \theta = \frac{\sqrt{7}}{4}$



(b)(i)  $FE = FB$  (tangents from external pt)  
 $\therefore \angle BEF = \angle FEB = \alpha$  (sides opposite equal angles)  
 $\angle AEB = \frac{\pi}{2}$  (angle in semi circle)

$\therefore \angle BED = \frac{\pi}{2}$  (straight line = 180°)

$\therefore \angle FED = \angle BED - \angle BEF$   
 $= \frac{\pi}{2} - \alpha$

(iii) In  $\triangle DEB$ ,  $\beta = \pi - \frac{\pi}{2} - \alpha$  (sum of angles)  
 $= \frac{\pi}{2} - \alpha$

$\therefore FD = FE$  (sides opposite equal angles)

But  $FE = FB$  (tangents from external pt)

$\therefore FD = FB$

(c)  $3 \cdot 2^2 + 3^2 \cdot 2^3 + \dots + 3^n \cdot 2^{n+1} = \frac{12}{5} (6^n - 1)$

Let  $n=1$   $\therefore$  LHS =  $3 \cdot 2^2$   
 $= 3 \times 4$   
 $= 12$

RHS =  $\frac{12}{5} (6 - 1)$   
 $= \frac{12}{5} \times 5$   
 $= 12$

$\therefore$  true for  $n=1$

### 14(c) (Continued)

Assume true for  $n=k$

$$\therefore 3 \cdot 2^2 + 3^2 \cdot 2^3 + \dots + 3^k \cdot 2^{k+1} = \frac{12}{5}(6^k - 1)$$

Prove true for  $n=k+1$

$$\text{ie Prove } 3 \cdot 2^2 + \dots + 3^k \cdot 2^{k+1} + 3^{k+1} \cdot 2^{k+2} = \frac{12}{5}(6^{k+1} - 1)$$

$$\text{LHS} = \frac{12}{5}(6^k - 1) + 3^{k+1} \cdot 2^{k+2}$$

$$= \frac{12}{5}(6^k - 1) + 3^k \cdot 3 \cdot 2^k \cdot 2^2$$

$$= \frac{12}{5}(6^k - 1 + \frac{12 \cdot 6^k}{5})$$

$$= \frac{12}{5}(6^k - 1 + 5 \cdot 6^k)$$

$$= \frac{12}{5}(6 \cdot 6^k - 1)$$

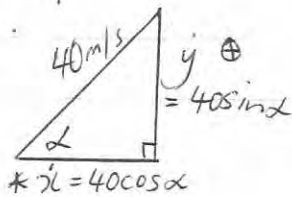
$$= \frac{12}{5}(6^{k+1} - 1)$$

$$= \text{RHS}$$

$\therefore$  If true for  $n=k$ , then also true for  $n=k+1$ . Is true for  $n=1$   $\therefore$  also for  $n=2$ ; for  $n=3$  etc for all positive integers  $n \geq 1$ .

(d) When  $t=0$ :

$$x=0, y=50$$



$$\frac{d^2x}{dt^2} = 0$$

$$\frac{d^2y}{dt^2} = -10$$

$$\frac{d}{dt} \frac{dy}{dt} = -10t + c$$

$$\frac{dy}{dt} = -10t + 40 \sin \alpha$$

$$y = \int (-10t + 40 \sin \alpha) dt$$

$$y = -5t^2 + 40t \sin \alpha + c$$

$$\text{When } t=0, y=50 \therefore c=50$$

$$\therefore y = -5t^2 + 40t \sin \alpha + 50$$

$$\frac{dx}{dt} = \int 0 dt = c$$

$$\frac{dx}{dt} = 40 \cos \alpha$$

$$\therefore x = \int 40 \cos \alpha dt$$

$$x = 40t \cos \alpha + c$$

$$\text{When } t=0, x=0 \therefore c=0$$

$$\therefore x = 40t \cos \alpha$$

### 14(d) (Continued)

(ii) When the projectile lands

$$y=0 \text{ and } x=200$$

$$\therefore -5t^2 + 40t \sin \alpha + 50 = 0 \quad (1)$$

$$\text{and } 40t \cos \alpha = 200 \quad (2)$$

$$\text{From (2)} \quad t = \frac{5}{\cos \alpha}$$

Sub. in (1):

$$-5 \left( \frac{25}{\cos^2 \alpha} \right) + 40 \sin \alpha \left( \frac{5}{\cos \alpha} \right) + 50 = 0$$

$$-125 \sec^2 \alpha + 200 \tan \alpha + 50 = 0$$

$$-125(1 + \tan^2 \alpha) + 200 \tan \alpha + 50 = 0$$

$$125 \tan^2 \alpha - 200 \tan \alpha + 75 = 0$$

$$5 \tan^2 \alpha - 8 \tan \alpha + 3 = 0$$

$$(5 \tan \alpha - 3)(\tan \alpha - 1) = 0$$

$$\therefore \tan \alpha = \frac{3}{5} \text{ or } \tan \alpha = 1$$

$$\alpha = 31^\circ \text{ or } \alpha = 45^\circ$$

(iii) Let  $\alpha = 45^\circ$

max  $y$  value when  $\frac{dy}{dt} = 0$

$$\therefore -10t + 40 \sin 45^\circ = 0$$

$$\frac{40}{\sqrt{2}} = 10t$$

$$t = \frac{4}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= 2\sqrt{2} \text{ seconds}$$

$$\therefore \text{max. } y = -5(2\sqrt{2})^2 + 40(2\sqrt{2}) \cdot \frac{1}{\sqrt{2}} +$$

$$= -40 + 80 + 50$$

$$= 90 \text{ metres.}$$